

DANTE LAMENZA NAYLOR

Brown Physics Department $\diamond$ Providence, RI
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EDUCATION

University of Massachusetts Amherst

Graduated May 2024

B.S. in Physics

B.S. in Applied Mathematics

Overall GPA: 3.9/4

Brown University

Expected May 2029

PhD in Physics

Overall GPA: 3.9/4

HONORS AND AWARDS

Brown Physics Elbaum Conference Travel Award
(\$700)

2025

UMass Amherst Rising Researcher Award ([link](#))

2023

Simenas Teaching Fellowship Award
(\$2,000)

2023

APS March Meeting Travel Award
(\$500)

2023

APS DFD Travel Award
(\$500)

2022

Chang Research Award from UMass Physics Department
(\$2,500)

2022

National Science Foundation grant #CHE-1659266
(\$5,500)

2021

MANUSCRIPTS

1. V. Mathai, A. Das, **D. L. Naylor**, K. S. Breuer,
“Shape-Morphing Dynamics of Soft Compliant Membranes for Drag and Turbulence Modulation”,
Phys. Rev. Lett. **131**, 114003 (2023) ([link](#))
2. **D. L. Naylor**, Z. Yang, R. O. Monico, E. Calzavarini, V. Mathai,
“Clustering Extremes for Bubbles and Particles in Decaying Isotropic Turbulence”,
(in preparation), (2026)

RESEARCH EXPERIENCE

Comparing turbine flow scales in varied atmospheric stabilities

Marston research group, Brown University

Fall 2025 - Present

Providence, RI

- Actively developing large eddy simulations of wind turbines in varying atmospheric-stability flows with PALM model system.
- Applying the Fokker-Planck operator to decompose the dynamic scales of wind turbine systems for further analysis.

Using Fokker-Planck Operator for wind farm forecasting

Marston research group, Brown University

Fall 2024 - Present

Providence, RI

- Developing predictions of wind farm systems by interpolating the Fokker-Planck operator in Julia.

Analysis of particles in decaying turbulence

Mathai research group, University of Massachusetts Amherst

Summer 2024 - Spring 2025

Amherst, MA

- Investigated Voronoi statistics and correlation dimension of point particles in self-decaying turbulence.
- Observed clusters tend to prefer interacting with eddies of their own size, resulting in nontrivial clustering effects.

Code development for membrane-fluid interaction

Mathai research group, University of Massachusetts Amherst

Summer 2022 - Spring 2023

Amherst, MA

- Implemented panel methods in MATLAB for membrane coupled with two-dimensional flow.
- Application to energy extraction in unsteady fluid flows.

Experimental research on fluidic coupled oscillators

Mathai research group, University of Massachusetts Amherst

Fall 2021 - Fall 2022

Amherst, MA

- Designed and programmed computer vision software to keep track of the free surface of a fluid in a video.
- CAD designed, 3D printed, programmed a stepper motor with an Arduino, and machined parts to be used in an experimental apparatus.

Experimental research on hysteresis in Taylor-Couette flow

Physics of Fluids Group, University of Twente

Fall 2023

Enschede, Netherlands

- Designed and performed experiments to transition between turbulent states.
- Observed experimentally induced flow fields using torque measurements and laser-doppler anemometry.

Summer research on “two-dimensional materials”

Yan research group, University of Massachusetts Amherst

Summer 2021

Amherst, MA

- Programmed and utilized a simulation for the optical contrast of atomically thin hexagonal boron nitride under different viewing conditions.
- Operated optical spectroscopy and an atomic force microscope to prove thickness and topology of samples.

CONFERENCE TALKS (AS PRESENTING AUTHOR)

1. “Statistics of Air Flow Past Wind Turbines From a Fokker Planck Operator”,
American Physical Society March Meeting,
(November 2025) ([link](#))
2. “Particle laden decaying turbulence”,
12th International Conference on Multiphase Flows,
(February 2025) ([link](#))
3. “Elastic membrane interacting with unsteady fluid flows”,
American Physical Society March Meeting,
(March 2023) ([link](#))
4. “Soft Fluidic Coupled Oscillators”,
75th Annual meeting of the American Physical Society (APS) Division of Fluid Dynamics,
(November 2022) ([link](#))
5. “Exfoliating, Detecting, and Classifying Ultra-Thin Hexagonal Boron Nitride”,
Summer Undergraduate Research Symposium at the University of Massachusetts Amherst,
(August 2021) ([link](#))

TECHNICAL STRENGTHS

Computer Languages	Python, MATLAB, Julia, C, FORTRAN, Mathematica
Platforms	Familiar with Windows, Mac OS, and Linux

LANGUAGES

English	Native language
Portuguese	Intermediate listener, speaker, reading, and writing
Italian	Novice listener, speaker, reading, and writing

REFERENCES

J. B. Marston

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Kenny Breuer

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